PROJECT AND TEAM INFORMATION

Project Title

EcoLife – Ecosystem Simulation and Analysis System

Student / Team Information

Team Name / Team #	*ENTER TEAM NAME*
Team Member 1 (Team Lead)	*Mishra Adarsh* *2510372901* *adsamishra@gmail.com* 
Team Member 2	*Gupta Unnati* *2510372161* *unnatigupta0020@gmail.com* 
Team Member 3	*Budakoti Rishabh* *25101201427* *rishabhbudakoti5@gmail.com* 
Team Member 4	*Dheeman Amritaanshu* *2512520011* *amritaanshu7015@gmail.com* 

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

-An ecosystem is a network of organisms and environmental conditions, not a group of independent living things.
-Plants, herbivores, carnivores, and birds depend on food, water, shelter, and other organisms in the food web.
-A single change in the ecosystem can affect several populations. For example, if drought reduces plant growth, herbivores may have less food. Over time, carnivores may also be affected because the prey population becomes smaller.
-Other factors such as weather, seasons, reproduction, death, and natural events can also change the ecosystem.
-Studying these situations directly in nature is difficult because real environmental conditions cannot be controlled safely for testing. EcoLife provides a safe virtual environment where users can observe these relationships over simulated days.
-The system allows comparisons between normal conditions, drought, forest fire, and predator-removal scenarios.
-EcoLife also provides a practical way to combine data structures, algorithms, and object-oriented programming in one connected application.

State of the Art / Current Solution (1 pt)

- Ecosystem behaviour is commonly studied through:
- Field observation
- Controlled experiments
- Mathematical models
- Computer simulations
- A simulation represents organisms, resources, and environmental conditions using defined rules.
- By applying these rules repeatedly, population changes and food-web interactions can be observed over time.
- Simulations allow researchers to study ecosystems without disturbing real habitats.
- Some scientific systems use large datasets and complex mathematical models for specialised ecological research.
- EcoLife takes a smaller and more educational approach.
- EcoLife does not claim to model a real ecosystem with complete scientific accuracy.
- Instead, it uses simple and understandable rules for:
- Plant growth
- Feeding
- Predation
- Reproduction
- Death
- Environmental events
- Users can advance the simulation by 1, 10, or 100 days.
- Users can observe the effects of:
- Drought
- Forest fire
- Predator removal
- This makes ecological interactions easier to understand and follow while demonstrating how connected factors affect populations.

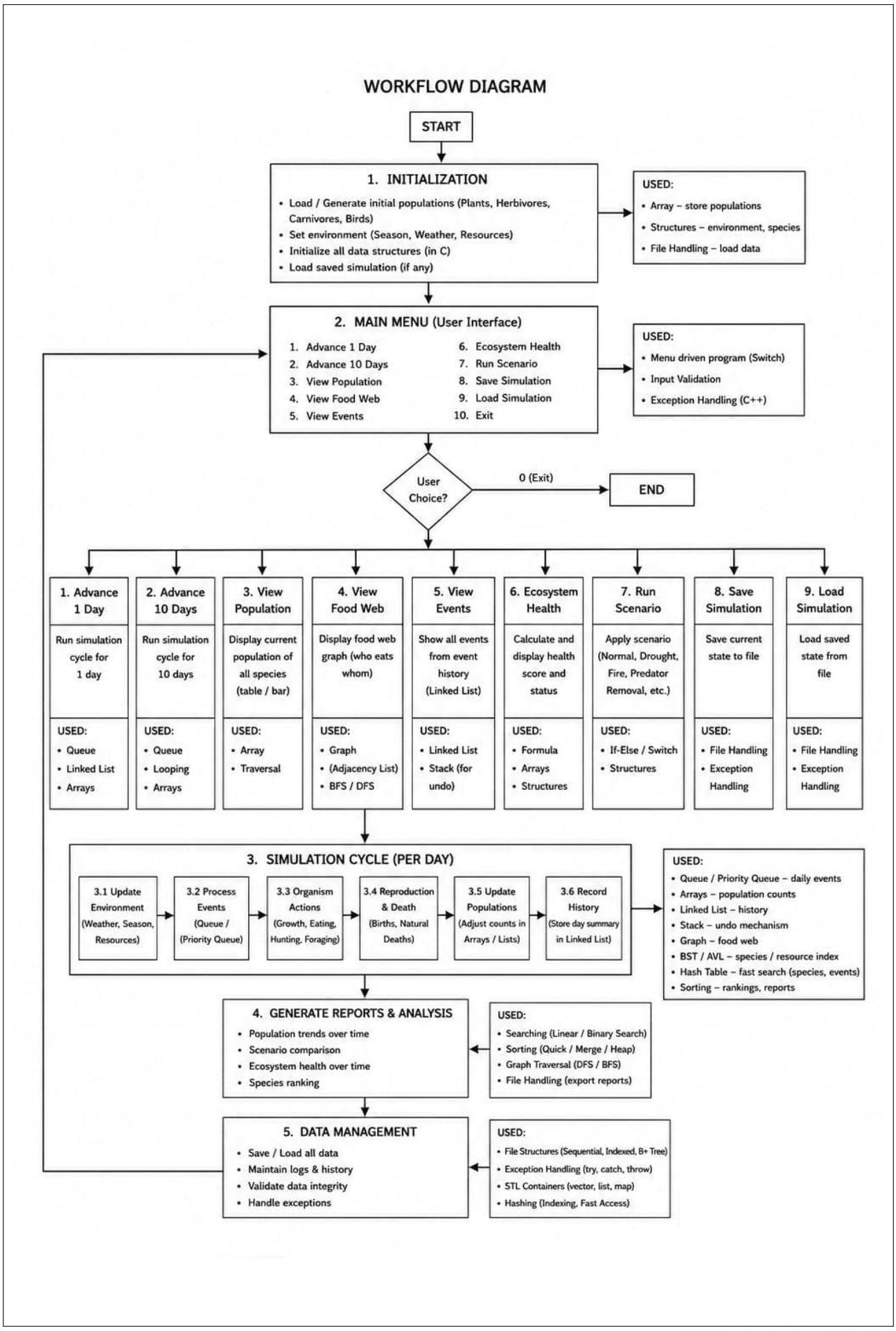
Project Goals and Milestones (2 pts)

- Main Goal of EcoLife: - Build a virtual ecosystem where organisms and environmental conditions influence each other over simulated time.
- Represent plants, herbivores, carnivores, and birds.
- Update populations based on:
- Food availability
- Feeding
- Predation
- Reproduction
- Death
- Include weather, seasons, and selected natural events. - Technologies and Concepts:
- Use C++ inheritance and polymorphism for the organism model.
- Use C data structures and algorithms for:
- Event management
- Organism lookup
- Population handling
- Food-web analysis

Project Approach (3 pts)

Programming Languages: - EcoLife will be developed using C and C++.
C++ – Ecosystem Modelling:
- The ecosystem will be represented using classes such as: - Organism - Plant - Animal - Herbivore - Carnivore - Bird - Environment - Weather - Season - Event - Ecosystem
- The Organism class will contain common properties and operations.
- Derived classes will define behaviour specific to plants and different animal types.
- Inheritance and polymorphism will allow different organisms to be handled through a shared design.
- This will keep the simulation organised and easier to expand.
C – Data Structures Algorithms:
- Arrays will organise population data. - Linked lists will manage dynamic organism collections. - Queues and priority queues will manage scheduled events. - Trees and hash tables will support efficient organism lookup. - Searching and sorting algorithms will help analyse population data. - The food web will be represented using a graph. - DFS and BFS will be used for food-web traversal and relationship analysis.
Daily Simulation Process:
- Update environmental conditions and resources. - Process scheduled events. - Apply relevant organism interactions. - Update population values. - Allow users to control and observe the simulation through a menu-based interface.

System Architecture (High Level Diagram) (2 pts)



Project Outcome / Deliverables (1 pt)

Final System:
- EcoLife will be a working, menu-based ecosystem simulation and analysis program. - It will combine:
- C++ for modelling organisms and their environment.
- C for data structures and algorithms.
Ecosystem Simulation:
- Run the ecosystem over virtual days.
- Update population values based on defined interaction and event rules.
User Features:
- View current population information.
- View event history.
- Explore food-web relationships.
- View a simple ecosystem-health value.
What-If Testing:
- Test the ecosystem under normal conditions.
- Simulate events such as:
- Drought
- Forest fire
- Predator removal
Final Submission:
- Source code
- Design report
- Presentation
- Test observations
- Working system demonstration

Assumptions

Purpose and Scope:
- EcoLife is designed as an educational simulation, not an exact scientific prediction of a real ecosystem.
- Population changes and environmental effects will follow predefined rules and simple probabilities.
Simulation Time:
- One virtual day will represent a fixed time step.
- The simulation will start with selected population values and environmental conditions.
Simplified Interactions:
- Feeding, predation, reproduction, and death will be simplified.
- This keeps the model easy to understand and manageable within the project scope.
Natural Events:
- Events such as drought and forest fire will have predefined effects on:
- Resources
- Organism populations
System Assumptions:
- The system assumes valid user input.
- It assumes sufficient memory is available for the selected population size.

References

- B. Stroustrup, *The C++ Programming Language*, 4th ed., Addison-Wesley Professional, 2013.
- B. W. Kernighan and D. M. Ritchie, *The C Programming Language*, 2nd ed., Prentice Hall, 1988.
- T. H. Cormen, C. E. Leiserson, R. L. Rivest and C. Stein, *Introduction to Algorithms*, 4th ed., MIT Press, 2022.
- OpenStax, *Biology 2e*, "46.1 Ecology of Ecosystems," Rice University.
- OpenStax, *Biology 2e*, "45.6 Community Ecology," Rice University.